BPM2_NB_ESOFT QUIZ 9_S2026

# of Questions: 30

Question #: 1

A 76-year-old man comes to the physician because of a 2-month history of heightened sensitivity to loud, low-pitched sounds. Which of the following cranial nerves is most likely involved?

A. Vestibulocochlear
✓B. Facial
C. Vagus
D. Glossopharyngeal
E. Hypoglossal

Rationale: The facial nerve provides innervation to the stapedius muscle which is responsible for dampening of sound by contracting and preventing the stapes from moving excessively.

Item Description: A 76-year-old man co

Question #: 2

A 40-year-old man is admitted to the hospital because of a fracture of the right pterion as a result of a car crash. A CT scan shows fluid buildup in the cranium. In which of the following areas is the hemorrhage most likely located?

A. Subdural space
B. Subarachnoid space
C. Lateral ventricle
✓D. Epidural space
E. Superior sagittal sinus

Rationale: Fracture of the pterion typically causes damage of the middle meningeal artery. Since it supplies the dura mater it travels between the dura and the bone in the epidural space.

Item Description: A 40-year-old man is

Question #: 3

A 20-year-old woman comes to the physician because of a 1-week history of bleeding gums when brushing her teeth. She reports missing her menstrual periods for the past 2 months. She has been on a very-low-calorie diet for the past 6 months because she believes she is overweight. Her BMI is 16.5 kg/m². Physical examination shows perifollicular hemorrhages and petechiae on the lower extremities. A diagnosis of a dietary vitamin deficiency secondary to anorexia nervosa is made. Which of the following processes is most likely defective due to the vitamin deficiency in this patient?

A. Formation of disulfide bonds in procollagen
B. Cross-linking of elastin monomers
✓C. Formation of hydroxyprolyl residues
D. Formation of allysine residues
E. Stability of desmosine
F. Synthesis of purine and pyrimidine nucleotides

Rationale: The vignette describes a patient with vitamin C deficiency. Vitamin C is required as a coenzyme for prolyl and lysyl hydroxylase. It facilitates the addition of -OH groups to proline, which in turn assists with the formation of Hydrogen bonds to stabilize the triple helical collagen structure.
Folic acid (Tetrahydrofolate) is required for purine and pyrimidine nucleotide synthesis.
Copper is required for the activity of lysyl oxidase, which is required for the formation of allysine and the formation of covalent cross-links in collagen.

Item Description: BCHM.0153; VitC deficiency

Question #: 4

A 3-week-old boy is brought to the emergency department by his parents due to a 2-day history of non-bilious, projectile vomiting after feeding. The parents report the infant has been increasingly fussy and has fewer wet diapers. Physical examination shows a palpable, firm, ovoid mass in the epigastric region and visible peristaltic waves across the upper abdomen. Imaging studies confirm pyloric stenosis, causing gastric outlet obstruction. Which of the following arterial blood gas (ABG) profiles is most likely in this patient?

A. I
B. II
C. III
✓D. IV
E. V

Rationale: Pyloric stenosis (gastric outlet syndrome) is characterized by metabolic alkalosis (loss of gastric HCl). Arterial pH is increased; Serum HCO3- is increased (loss of H+ in gastric fluid adds a bicarbonate to the ECF); PCO2 is elevated as a compensatory response (alkalosis inhibits the respiratory center and results in hypoventilation)

Attachment:

	pH	PCO_2	HCO_3^-
I	Normal	Increased	Decreased
II	Increased	Decreased	Increased
III	Decreased	Increased	Normal
IV	Increased	Increased	Increased
V	Decreased	Decreased	Normal

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Item Description: Pyloric stenosis - quiz

Question #: 5

A 32-year-old audio engineer comes to the physician because of a 2-month history difficulty distinguishing high-frequency sounds after prolonged exposure to loud studio monitors. Physical examination of the external ear and otoscopic examination are normal. Audiometry shows signs of early sensorineural hearing loss. Damage to the cells responsible for the conversion of mechanical vibrations into nerve impulses within the organ of Corti, is suspected. Which of the following most likely represents the damaged structures?

✓A. Outer and inner hair cells with stereocilia
B. Type I and type II hair cells attached to a cupula
C. Type I and type II hair cells with a true kinocilium
D. Inner and outer phalangeal cells
E. Inner and outer pillar cells

Rationale: Inner and outer hair cells are the primary sensory receptors of the organ of Corti.

They contain stereocilia that bend in response to vibration-induced movement of the basilar membrane.
This deflection opens mechanically gated ion channels, initiating the mechanoelectrical transduction that ultimately generates auditory nerve signals/impulses.
These cells are essential for normal hearing and are highly susceptible to damage from chronic loud noise.

Item Description: HCB.1384 - Organ of Corti structure

Question #: 6

A 32-year-old woman presents for a routine eye examination. She has no visual complaints, and her visual acuity is 20/20 bilaterally. Funduscopic examination shows a normal-appearing optic disc with a distinct margin, a physiologic cup, and normal retinal vessels emerging from the center. This region corresponds to the point of the retina that creates a natural blind spot. Which of the following best describes the normal histology of this part of the retina?

A. It contains a high density of photoreceptors that transmit visual information to the optic nerve.
✓B. It is composed primarily of unmyelinated retinal ganglion cell axons that become myelinated only after passing through the lamina cribrosa.
C. It contains the highest concentration of bipolar cells in the retina.
D. It is lined by a single layer of retinal pigment epithelium that overlies choroidal vasculature.
E. It contains multiple layers of Müller cells forming the physiologic cup.

Rationale: The optic disc is the point where retinal ganglion cell axons converge and exit the eye to form the optic nerve.
Within the globe, these axons are unmyelinated.

They acquire myelin only after passing through the lamina cribrosa, where they transition from the intraocular to the retrobulbar optic nerve.
The optic disc lacks photoreceptors, creating the natural blind spot.

Item Description: HCB.1346 - Retina; optic disc

Question #: 7

A 34-year-old man comes to the physician because of a 4-year history of episodic vertigo, nausea and vomiting. Neurological examination shows rhythmic oscillation of the eyes, with slow leftward drift followed by fast resetting to the right. A diagnosis of Meniere's disease is made. Which of the following pathological conditions most likely triggers the episodes of dizziness and related events?

A. Hemorrhage of the dorsolateral medulla
B. Schwannoma of CN VIII
C. Inflammatory compression of CN VII
✓D. Labyrinthine endolymphatic dysfunction
E. Intracranial infection

Rationale: Meniere's disease can affect either or both the vestibular and cochlear systems. It is more commonly unilaterally expressed. The condition reflects anomalies pf the endolymphatic spaces, possibly contributing to mechanical or chemical pathologies.

Item Description: SOM.CS.II.BPM2.4.NB.4.NSCI.0443 Menier's disease

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Question #: 8

A 23-year-old man is brought to the emergency department by his friend because of a 1-hour history of trauma to the head. He is presently unconscious. Neurologic examination shows left hyperreflexia. CT studies show blood following the sulci. Which of the following vascular structures has most likely ruptured?

A. Bridging veins
✓B. Middle cerebral artery
C. Middle meningeal artery

D. Inferior sagittal sinus
E. External carotid artery

Rationale: The hemorrhage is of subarachnoid origin, given that the accumulating blood follows the convolutions of the brain. The only artery listed that dwells within the subarachnoid space is the middle cerebral artery.

Item Description: SOM.CS.II.BPM2.4.NB.4.NSCI.0424 Elevated intracranial pressure

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Question #: 9

A 68-year old man comes to the physician because of a 6-month history of worsening vision. Examination of the retina in an experimental protocol shows optic atrophy with preserved photosensitive retinal ganglion cells. Which of the following diencephalic nuclei is most likely to show normal rhythmic functions?

A. Lateral geniculate
B. Medial geniculate
C. Ventral posterolateral
✓D. Paraventricular
E. Pulvinar

Rationale: Preservation of photosensitive retinal ganglion cells means that the suprachiasmatic nucleus is being normally stimulated by sunlight. The suprachiasmatic nucleus inhibits the paraventricular nucleus, which leads to inhibition of melatonin production. Thus, the paraventricular nucleus should show a normal rhythmic function. The LGN, MGN, and VPLN are sensory relay neurons of the thalamus and are not directly innervated by the photosensitive retinal ganglion cells. The pulvinar is a higher-order thalamic nucleus involved in attention but is not directly innervated by the suprachiasmatic nucleus in a rhythmic manner.

Item Description: SOM.MK.I.BPM2.4.NB.1.NSCI.0300 Suprachiasmatic nucleus

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Question #: 10

A 5-year-old boy is brought to the physician by his mother because of a 3-month history of sleep-related concerns. He regularly sleeps 14 hours nightly. Which of the following descriptions best applies to his quantity of sleep?

A. Insufficient for his age
B. Recommended for his age
✓C. Appropriate for his age
D. Too much for his age

Rationale: According to U.S. National Sleep Foundation standards, 14 hours a night is within the appropriate range for a 5-year-old child.

Item Description: SOM.MK.IV.BPM2.4.NB.1.NSCI.0307 Sleep deprivation

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Question #: 11

A 36-year-old woman is brought to the emergency room because of a 30-minute history of traumatic injury and respiratory distress. She is intubated due to ataxic breathing. Which of the following brain regions is most likely damaged?

A. Pontine respiratory group
B. Apneustic center
✓C. Ventral medullary respiratory group
D. Midbrain
E. Phrenic motor nucleus

Rationale: Ataxic breathing is associated with damage to the primary respiratory rhythm generators in the ventral medulla, mainly the pre-Bötzinger complex and Bötzinger complex.

Item Description: SOM.MK.I.BPM2.4.NB.1.NSCI.0298 Respirator modulation

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Question #: 12

An 81-year-old man comes to the physician because of a 10-year history of progressively declining duration of sleep. He says that he previous slept between 7 and 8 hours per night. He now sleeps between 5 and 6 hours per night. It is determined that the declining durations of sleep are age-related. Age-related cellular loss from which of the following nuclei is most likely implicated in his altered pattern of sleep?

A. Superior colliculus
B. Paraventricular
✓C. Ventral lateral preoptic intermediate
D. Lateral geniculate
E. Paramediane pontine reticular formation

Rationale: Diminished sleep duration associated with aging relates to the loss of galanin-producing cells in the VLPO nucleus of the hypothalamus.

Item Description: SOM.MK.I.BPM2.4.NB.1.NSCI.0306 Age-relate changes in sleep

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Question #: 13

A 19-year-old man is brought to the physician by his parents because of a 3-month history of dramatic behavioral changes that have resulted in legal problems. He has been arrested for shoplifting, driving under the influence of alcohol, and possession of cocaine. These behavioral problems began after he started college. He says he feels overwhelmed by being away from home and by the academic pressures of school. Which of the following is the most likely diagnosis?

A. Acute stress disorder
B. Posttraumatic stress disorder
✓C. Adjustment disorder with disturbance of conduct
D. Adjustment disorder with anxiety
E. Adjustment disorder with depressed mood

Rationale: This person has a maladaptive response to an identifiable stressor that is not explained by any other diagnosis. The person has not experienced a "traumatic" stressor nor the required set of symptoms (intrusion, avoidance, negative change in mood/thoughts, and hyperarousal) for a diagnosis of PTSD or ASD.   When the reaction in adjustment disorder pertains to misbehavior/misconduct, the subtype is "with disturbance in conduct."

Item Description: Esoft.quiz.BEHS.0079.trauma-stressor disorders_adjustment_diagnosis

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Question #: 14

A 23-year-old man comes to the physician because of a 2-year history of anxiety. He is distressed about a recurrent sexually violent image that comes to mind and performs a ritual to get rid of the image, but the image returns a few hours later. He is not interested in behavioral therapy. Which of the following medications is most appropriate for the physician to prescribe?

✓A. Paroxetine
B. Diazepam
C. Risperidone
D. Lithium
E. Selegiline

Rationale: This man has obsessive-compulsive disorder. Given that he is unwilling to undergo behavioral therapy, he most likely was started on an SSRI antidepressant, such as paroxetine. Benzodiazepines (diazepam), antipsychotics (risperidone), mood stabilizers (lithium) and MAOIs (selegiline) are not typically used for OCD.

Item Description: Esoft.quiz.BEHS.0072.OCD-related_OCD_treatment_SSRI

Question #: 15

A 25-year-old woman comes to the physician because of a life-long history of fear of the dark. She goes to great lengths to avoid being in the dark and experiences pronounced anxiety at the thought of being in such situations. She knows there is nothing to be afraid of in the dark, but this awareness does not reduce her fear. Which of the following therapeutic interventions is most appropriate?

A. Aversion therapy
B. Psychodynamic therapy
✓C. Exposure therapy
D. Covert sensitization
E. Cognitive restructuring

Rationale: The best answer here is exposure therapy, which is based on classical extinction. Exposure therapy involves gradually exposing the woman (either imaginal or in vivo) to the feared stimulus (being in the dark) and having her remain in the presence of the feared stimulus until there is some lessening of the anxiety.

Cognitive restructuring involves changing one's cognitive biases (e.g., negative filter). In this case, the woman already "knows" that there is nothing to fear about the dark but fears it nonetheless. Cognitive therapy by itself would not be the best choice to treat her fear.

Item Description: ESoft.quiz.BEHS.0069.anxiety disorders_specific phobia_treatment

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Question #: 16

A 45-year-old man comes to the physician because of a 9-month history of excessive daytime sleepiness. He takes irresistible naps multiple times each day. He frequently loses muscle tone when very emotional and falls to the ground while retaining consciousness. Which of the following treatments is most appropriate?

A. Antidepressants
✓B. A stimulant and an antidepressant medication
C. Antipsychotics
D. Lithium
E. Stimulants
F. Sleep hygiene techniques

Rationale: The person's symptoms of irresistible sleep and cataplexy suggest narcolepsy. The two types of symptoms can be treated separately-- stimulants for the sleepiness and antidepressant medication to help cataplexy by suppressing manifestations of REM sleep (recall that cataplexy is viewed as the occurrence of REM-sleep paralysis during wakefulness).

Item Description: Esoft.quiz.BEHS.0096.sleep-wake disorders_narcolepsy_treatment

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Question #: 17

A 29-year-old man is brought to the emergency department by his wife because of a 4-hour history of headache, nausea, and vomiting. These symptoms started after eating pepperoni pizza and drinking an alcoholic beverage at a party. He has been taking medication for the past 6 months for a psychiatric condition. Physical examination shows hypertension, tachycardia, and profuse sweating. Which of the following medications most likely accounts for his symptoms?

A. Selective serotonin reuptake inhibitor
B. Serotonin-norepinephrine reuptake inhibitor
✓C. Monoamine oxidase inhibitor
D. Tricyclic antidepressant
E. Antipsychotic
F. Benzodiazepine

Rationale: Alcohol and pepperoni contain tyramine, which is not metabolized when a person takes an MAOI. The tyramine buildup can cause a hypertensive crisis, characterized by the physical symptoms described in this scenario.

Item Description: Esoft.quiz.BEHS.0087.depressive disorders_treatment_MAOIs

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Question #: 18

A 23-year-old woman comes to the physician because of a 3-month history of a major depressive episode. She is prescribed an antidepressant medication that selectively inhibits the reuptake of serotonin. Which of the following medications was she most likely prescribed?

A. Phenelzine
B. Imipramine
C. Diazepam
✓D. Fluoxetine
E. Venlafaxine

Rationale: The question is asking about an SSRI. Fluoxetine is the only SSRI listed.
Phenelzine is a MAOI (monoamine oxidase inhibitor); imipramine is a TCA (tricyclic antidepressant); diazepam is a benzodiazepine; venlafaxine is a SNRI (serotonin and norepinephrine reuptake inhibitor)

Item Description: Esoft.quiz.BEHS.0087.depressive disorders_treatment_SSRIs

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Question #: 19

A 30-year-old man comes to the physician because of a 7-month history of fear that he has multiple sclerosis. He overinterprets normal bodily symptoms such as feeling tightness in his legs after exercising and having his vision momentarily blurred when changing his visual focus from a close object to a distant target. Physical examination, laboratory studies, and MRI of the brain show no abnormalities. Despite these reassurances, he remains fearful and preoccupied with the thought that he might have multiple sclerosis. Which of the following is the most likely diagnosis?

A. Somatic symptom disorder
B. Conversion disorder (functional neurological symptom disorder)
✓C. Illness anxiety disorder
D. Delusional disorder, somatic type
E. Body dysmorphic disorder

Rationale: The person has excessive thoughts and behaviors about normal physical sensations. He does not present with significant physical complaints. Excessive worry about health in the context of not having significant symptoms is the basis for a diagnosis of illness anxiety disorder. Conversion requires a pseudoneurological symptom (e.g., deafness, blindness, seizures) with evidence that the underlying neurological system is intact. Delusional disorder requires that the patient be 100% certain of a belief, regardless of the evidence.

Item Description: Esoft.quiz.BEHS.0074.somatic symptom disorders_IAD_diagnosis

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Question #: 20

A 25-year-old woman comes to the physician because of a 4-month history of depressed mood, low energy, difficulty sleeping, lack of appetite, feelings of worthlessness, and anhedonia. Her symptoms are severe, and she is unable to work. This is her third time experiencing this type of mood episode over the past 2 years. She also has a history of episodes characterized by distinctly elevated mood with increased energy, pressured speech, and a reduced need for sleep. Those episodes last 3-4 months, and she initiates and completes many new projects in a brief time without any compromise in her functioning during those times. Which of the following is the most likely diagnosis?

A. Bipolar I disorder
B. Cyclothymic disorder
C. Persistent depressive disorder
✓D. Bipolar II disorder
E. Major depressive disorder

Rationale: This woman has lows and highs in her mood. The lows are significant, and she has sufficient symptoms for an MDE. On the other hand, her high periods do not cause significant compromise/ impairment. Thus, her symptoms during her elevated mood states are more compatible with HYPOmania. Recall that the symptoms of mania and hypomania are identical; the major difference is the degree of functional impairment experienced (mania is significantly compromising). In this case, she experiences MDEs and hypomanic episodes. This combination of mood episodes is Bipolar II disorder.

Item Description: Esoft.quiz.BEHS.0089.bipolar disorders_bipolar II_diagnosis

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Question #: 21

A 10-year-old boy is brought to the physician by his parents because of a 9-month history of disturbed sleep. He was born with cerebral palsy and walks using orthotics to improve his balance. Within the first 90 minutes of falling asleep, he screams and wakes up terrified. He is unable to explain what frightens him and does not remember the episode the next morning. He also has episodes of sleep walking, which poses a risk of falls given his unstable gait. The physician prescribes a medication that improves both sleep-related behaviors. Which of the following changes to his sleep architecture is most likely responsible for the improvement in his symptoms?

A. Increased slow wave sleep
✓B. Decreased slow wave sleep
C. Increased REM sleep
D. Decreased REM sleep
E. Decreased Stage 2 sleep

Rationale: Based on the lack of dream imagery, amnesia for the event, and timing of the sleep event (early in the sleep episode), the boy most likely experiences sleep terrors. Terrors are a slow-wave sleep phenomenon (stage N3), so medication to treat terrors would most likely suppress this stage of sleep.

Item Description: Esoft.quiz.BEHS.0096.sleep-wake disorders_sleep terrors_treatment_effect on architecture

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Question #: 22

A 45-year-old man comes to the physician because of excessive tiredness during the day. He sleeps at least 10 hours per night and takes naps, yet never feels alert. Polysomnography shows no abnormalities other than extended total sleep time. His BMI is 24 kg/m2. Which of the following treatments is most likely recommended?

A. Continuous positive airway pressure
B. Improvement in sleep hygiene
✓C. Stimulant medication
D. Tricyclic antidepressants
E. Benzodiazepines

Rationale: The person experiences excessive daytime sleepiness despite adequate sleep and a normal PSG (other than duration of sleep). This is most likely hypersomnia, which is most effectively managed with a stimulant or wake-promoting medication.

Item Description: Esoft.quiz.BEHS.0096.hypersomnia_treatment

Question #: 23

A 48-year-old woman comes to the physician because of a 4-month history of headaches and blackouts. These symptoms started after slipping on a wet restaurant floor and hitting her head. She says she was functioning "perfectly" in life until the fall and now she is "totally debilitated." She is suing the restaurant for a large sum of money. Consultation with family members shows that she has been facing significant occupational, social, and financial difficulties for the past year. Results of extensive evaluations show no abnormalities. Which of the following is the most likely diagnosis?

A. Illness anxiety disorder
✓B. Malingering
C. Factitious disorder, imposed on self

D. Factitious disorder, imposed on another
E. Conversion disorder (functional neurological symptom disorder)
F. Malingering by proxy

Rationale: In this scenario, there is evidence that the woman is lying about her financial state. At the same time, she is claiming neurological problems when evidence suggests that her neurological system is intact. Given the lawsuit involved and the history of lying about her financial state, it appears there is a strong financial motivation to be lying about her problems. Malingering is the most likely explanation for her behavior.

Item Description: Esoft.quiz.BEHS.0074.somatic symptom disorders_malingering

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Question #: 24

A 25-year-old woman comes to the physician because of a 6-month history of a painful lump in her neck. Laboratory tests and physical examination show that the lump is a swollen lymph node from a viral infection of the upper respiratory tract. Despite the reassurances, she is fearful that a more serious condition has been overlooked. Because of the discomfort in her neck and her excessive fear about having a serious disease, she is unable to work. Which of the following is the most likely diagnosis?

✓A. Somatic symptom disorder
B. Conversion disorder (functional neurological symptom disorder)
C. Illness anxiety disorder
D. Delusional disorder, somatic type
E. Body dysmorphic disorder

Rationale: This person experiences a genuinely distressful physical symptom. She is worried (not 100% convinced) that she has a more serious condition, and her concern is excessive and impairing to her life. This fits SSD. In IAD, there are no significant physical symptoms. In delusional disorder, the person is 100% convinced of a belief without a basis.

Item Description: Esoft.quiz.BEHS.0074.somatic symptom disorders_SSD_diagnosis

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Question #: 25

A 50-year-old man comes to the physician because of a life-long history of mood lability. For several months each year, he feels depressed, lacks self-esteem, and has low energy. For the remaining months of the year, his mood is elevated, and he feels full of energy, sleeps little, talks in a pressured manner, and starts new projects. His fluctuations in mood have not significantly compromised his functioning. Which of the following is the most likely diagnosis?

A. Bipolar I disorder
✓B. Cyclothymic disorder
C. Persistent depressive disorder
D. Bipolar II disorder
E. Major depressive disorder

Rationale: The person experiences observable ups and downs in his mood, but his symptoms don't ever meet the requirements for either an MDE or a manic episode. Thus, the only diagnosis that is possible to capture these chronic oscillations in mood is cyclothymia.

Item Description: Esoft.quiz.BEHS.0089.bipolar disorders_cyclothymia_diagnosis

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Question #: 26

A 48-year-old man is brought to the physician by a co-worker because of a 1-month history of complete autobiographical memory loss. He has no memories of who he is or anything about his personal life. The physician discovers that the patient is a missing person from a neighboring town who disappeared after his wife was killed in a car crash. Which of the following is the most likely diagnosis?

A. Dissociative amnesia (localized)
✓B. Dissociative amnesia with dissociative fugue
C. Dissociative amnesia (selective)
D. Posttraumatic stress disorder
E. Depersonalization/derealization disorder
F. Dissociative identity disorder
G. Acute stress disorder

Rationale: The man has memory loss due to an apparent psychosocial stressor AND travels away. This makes the answer dissociative amnesia with dissociative fugue. If he had not traveled away, the answer would be dissociative amnesia (generalized).

Item Description: Esoft.quiz.BEHS.0082.dissociative disorders_amnesia with fugue_diagnosis

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Question #: 27

A 21-year-old man comes to the physician because of a 6-week history of emotional difficulties. These difficulties started within 1 week of returning from military duty in which he witnessed deadly combat. He experiences insomnia almost every night and when he is able to sleep, he has nightmares about the war. He stays in his room all day and does not return calls from friends or family. He avoids anything that reminds him of being in war. He feels anhedonic, detached, and guilty and is prone to angry outbursts without provocation. Which of the following is the most likely diagnosis?

A. Acute stress disorder
✓B. Posttraumatic stress disorder
C. Adjustment disorder with anxiety
D. Generalized anxiety disorder
E. Social anxiety disorder
F. Social anxiety disorder, performance only

Rationale: This person has at least 1 symptom in each of the 4 major categories of PTSD symptoms (intrusions, avoidance, negative change in mood/cognition, hyperarousal). The symptoms have persisted for more than a month, so the diagnosis is PTSD.

Item Description: Esoft.quiz.BEHS.0079.trauma-stressor disorders_PTSD_diagnosis

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Question #: 28

A 34-year-old woman comes to the physician because of a 2-year history of fear, anxiety, and avoidance of public speaking. She is a research scientist and says she becomes paralyzed with fear when speaking in front of a crowd. She says, "My heart pounds, my hands shake, and my mouth gets so dry that I cannot speak." Her fear and anxiety about public speaking relate to excessive concerns about being negatively evaluated by her peers. She now rarely speaks at scientific conferences, which has reduced the circulation of her scientific publications. Which of the following is the most likely diagnosis?

A. Agoraphobia
B. Agoraphobia and Panic disorder
C. Specific phobia, situational
✓D. Social anxiety disorder, performance only
E. Panic disorder
F. Generalized anxiety disorder

Rationale: This woman experiences intense fear and avoidance behavior in response to a specific trigger (a social setting) and specific to one social setting (public speaking). Her fear/anxiety/avoidance behavior in this social setting relates to her fear of negative evaluation. This is the essence of social anxiety disorder, performance only. Since the person meets the criteria for social anxiety disorder, then the diagnosis of specific phobia is not necessary.

In panic disorder, the episodes of anxiety occur in "random" (uncued) situations. Agoraphobia refers to a fear of being in multiple difficult to escape from places (e.g., public transportation, wide-open spaces, enclosed spaces) if something unwanted happened.
Generalized anxiety disorder is constant worry about many different (and known) thoughts (i.e., a chronic worrier).

Item Description: ESoft.Quiz.BEHS.0067.anxiety disorders_social anxiety_diagnosis

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Question #: 29

A 56-year-old man comes to the physician because of a 1-year history of heartburn and abdominal discomfort. He has been taking antacids for the past 2 years. He has smoked 1 pack of cigarettes daily for 30 years. His BMI is 32 kg/m_2 (Normal: 19-25 kg/m^2). Physical examination shows no abnormalities. Endoscopy of the distal esophagus shows ulceration. Histologic examination of a biopsy of the mucosa in this region shows columnar epithelium with goblet cells. He is diagnosed with chronic gastroesophageal reflux disease with complications. Which of the following defects is most likely the cause of this patient's condition?

A. Impaired upper esophageal sphincter (UES) function
B. Absence of peristalsis in the distal esophagus
✓C. Impaired lower esophageal sphincter (LES) function
D. Defective functioning of the swallowing center
E. Defect in generation of primary peristaltic waves in the esophagus

Rationale: The correct answer is C. Impaired lower esophageal sphincter (LES) function results in chronic gastroesophageal reflux.

Item Description: PHYS.1103 GERD Etiology

Question #: 30

A 45-year-old man comes to the physician because of a 2-month history of headaches, muscle cramps, and fatigue. He also has difficulty keeping balance and feels that his muscles are weak. His blood pressure is 160/110 mm Hg; rest of the vital signs are within normal limits. Physical examination does not show any edema on ankles or knees. Laboratory studies show:

Test	Patient	Reference range
Serum Sodium	149 mEq/L	Normal: 135-145 mEq/L
Serum potassium	2.4 mEq/L	Normal: 3.5-5 mEq/L
Serum bicarbonate	45 mEq/L	Normal: 24-28 mEq/L
Plasma aldosterone	200 ng/dL	Normal: 10-30 ng/dL

Reduced serum concentration of which of the following hormones is most likely in this patient?

A. Catecholamines
B. Dehydroepiandrosterone
✓C. Renin
D. ACTH
E. Cortisol

Rationale: The plasma electrolytes along with the other symptoms of the person indicate that the patient might be having hyperaldosteronism (high plasma Na^+ , low K^+, and high HCO_3^- in addition to hypertension). To differentiate between the primary and secondary hyperaldosteronism plasma concentration of renin would be critical. Plasma renin is generally decreased in primary hyperaldosteronism but increased in secondary hyperaldosteronism. In both hyperaldosteronisms, plasma concentration of aldosterone is increased leading to the observed electrolyte imbalances as indicated in the plasma electrolyte analysis. Although catecholamines may cause hypertension of this patient but the serum electrolytes imbalances cannot be explained faithfully enough by this and thus plasma analysis of catecholamine is not necessary. Therefore, option A is incorrect. Option B is incorrect as dehydroepiandrosterone predominantly has androgenic effects and signs and symptoms of this patient cannot be explained rationally by this hormone. Option D and E are incorrect as these two hormones predominantly are not involved in the function of mineralocorticoid activity although a physiological level of ACTH is required for the proper function of aldosterone.

Item Description: PHYS.1029 Primary hyperaldosteronism Laboratory findings